DC Fast Charger Controller – Complete Technical Explanation

This topic covers four important areas of EV (Electric Vehicle) charging technology:

1. DC Fast Charger Controller
2. EV Communication Protocols
3. Charger-Side Hardware
4. Buck/Boost Converters

These are closely connected topics used in EV charging stations, power electronics, embedded systems and EV charger development.

Since you're interested in embedded systems and hardware testing, this is a useful technical domain to understand, especially for EV charger controller and hardware validation interviews.

1. DC Fast Charger Controller

Eaton Green Motion DC EV Charger 3-phase 50kW 480V
Factory Supply DC EV Charger Controller of DC Fast Charging Station Control System - DC EV Charger Controller and DC Fast EV Charger Controller
UR100020-SW充电模块_优优绿能
11

A DC Fast Charger (DCFC) supplies DC electrical power directly to an EV battery.

Unlike an AC home charger, which supplies AC power and relies on the vehicle's onboard charger to convert it into DC, a DC fast charger performs the main AC-to-DC conversion inside the charging station.

Basic working architecture:

Webinar: Fast EV Charging with onsemi: Explore the 25kW EliteSiC Module Based Reference Design - Discrete Semiconductor Products - DigiKey TechForum - An Electronic Component and Engineering Solution Forum
forum.digikey.com

1. Grid supplies AC electricity.
2. AC-DC rectifier converts AC into DC.
3. Power factor correction (PFC) improves input power quality.
4. DC-DC converter adjusts voltage and current.
5. Controller manages the charging process.

6. DC power reaches the EV battery through the charging connector.

The controller monitors voltage, current, temperature, safety conditions and communication with the vehicle.

2. EV Communication Protocols

[European EV Charging Standards Explained]
[Building an Interoperability Test System for Electric Vehicle Chargers Based on ISO/IEC 15118 and IEC 61850 Standards]
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7

The charger and EV must communicate before and during charging.

Protocol	Purpose
CCS / CCS2	DC charging connector and charging interface
CAN	Communication between electronic controllers
DIN 70121	Digital communication for DC charging
ISO 15118	Advanced EV-to-charger communication
CHAdeMO	DC charging standard
OCPP	Communication between charger and charging management backend

Important interview question: How does a charger communicate with an EV?

Example using CCS2:

- Vehicle connects to the charger.
- Charger detects the connection.
- Safety checks and connector locking take place.
- Communication is established using the applicable charging protocol.
- Vehicle BMS communicates battery voltage and charging requirements.
- Charger adjusts its output voltage and current.
- Charging stops safely when requested or when a fault occurs.

Remember: OCPP is primarily charger-to-backend communication, not the direct EV-to-charger protocol.

3. Charger-Side Hardware

VerdeMobility
เอาชนะความท้าทายด้านการออกแบบสำหรับการชาร์จ EV | DigiKey
CXT-PLA3SA12550A - SiC MOSFET Intelligent Power Module 1200V/550A, 3-phase, AlSiC PinFin heatsink
12

These are the major hardware components you should understand.

Component	Function
AC input circuit breaker	Protects against input faults
EMI filter	Reduces electromagnetic interference
Rectifier	Converts AC into DC
PFC circuit	Improves power factor
DC-link capacitor	Stabilizes DC voltage

SiC MOSFET / IGBT	High-speed power switching
Gate driver	Controls power semiconductor switching
DC-DC converter	Controls charging voltage/current
Contactor	Connects or disconnects high-voltage power
Current sensor	Measures charging current
Voltage sensor	Monitors voltage
Temperature sensor	Monitors overheating
MCU / DSP	Controls power electronics
PLC communication module	Communicates with the EV using CCS
Cooling system	Removes heat from power electronics

What would you do in a charger hardware testing role?

- Check PCB components and soldering quality.
- Perform visual inspection of connectors and wiring.
- Verify sensor readings.
- Test contactor operation and feedback.
- Monitor DC output voltage and current.
- Check temperature sensors and cooling fans.
- Inspect fault logs and communication errors.

- Validate emergency-stop and protection responses.

These are typical validation activities; actual procedures depend on the manufacturer's test plan and safety standards.

4. Buck and Boost Converters

What is Buck Converter ? Working and Circuit - Nerds Do Stuff
DC/DC Converters: Devices for Converting to a Higher Voltage - Technical Articles
Selecting a MOSFET for Buck-Boost DC-DC Converters in Car USB PD Designs - Industry Articles
6

This is an especially important topic in EV charger interviews.

A. Buck converter – Step-down

A buck converter reduces DC voltage.

Example:

- Input: 400V DC
- Output: 300V DC
- Application: Reducing DC voltage to match a battery's charging requirements.

B. Boost converter – Step-up

A boost converter increases DC voltage.

Example:

- Input: 300V DC
- Output: 600V DC
- Application: Increasing DC-link voltage in suitable power-conversion architectures.

C. Buck-Boost converter

A buck-boost converter can increase or decrease DC voltage.

A Review of Control Strategies for Four-Switch Buck–Boost Converters
mdpi.com

For example:

- Input voltage: 400V
- Required output: 350V → Buck operation.
- Required output: 600V → Boost operation.

The actual converter topology used in a commercial fast charger may be an isolated or non-isolated design, depending on the system architecture.

Important formula

For an ideal basic buck converter operating in continuous conduction mode:

$$ V_{out}=D\times V_{in} $$

For an ideal basic boost converter:

$$ V_{out}=\frac{V_{in}}{1-D} $$

Where D is the switching duty cycle (between 0 and 1).

For example, a buck converter with 400V input and 75% duty cycle ideally produces:

$$ 400\times0.75=300V $$

Real circuits have switching losses, conduction losses and control limitations.

5. Complete architecture – How everything works together

PowerFlex Doc Hub | docs.powerflex.com

Section	Main responsibility
AC-DC stage	Converts grid AC into DC
DC-link	Provides stable intermediate DC
DC-DC stage	Regulates battery charging voltage/current
Charger controller	Coordinates operation and protection

PLC controller	Handles vehicle communication
Sensors	Provide real-time feedback
Contactors	Control high-voltage connection
BMS	Reports battery charging requirements

A typical DC fast charger uses separate control functions for the power converter, vehicle communication and supervisory management.

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6. Important Interview Questions

Here are 15 questions to prepare for an EV charger controller or hardware engineering interview.

No.	Interview question
1	What is a DC Fast Charger?
2	What is the difference between AC and DC charging?
3	What is the function of a DC charger controller?
4	What is the difference between CAN and PLC communication?

5 What are CCS1 and CCS2 connectors?

6 What is ISO 15118?

7 What is the purpose of OCPP?

8 What is the function of a contactor?

9 What is a DC-link capacitor?

10 What is the difference between MOSFET and IGBT?

11 What is a gate driver?

12 What is the difference between buck and boost converters?

13 How do you detect overcurrent and overvoltage?

14 What happens when a charger overheats?

15 How would you troubleshoot a DC fast charger that is not charging?

Three questions you should answer confidently

Q1. What is the difference between CAN and PLC?

- CAN: A wired communication bus commonly used between internal electronic controllers.
- PLC: Power Line Communication, which carries digital communication over the charging cable's control-pilot path in CCS.

Q2. What is the difference between ISO 15118 and OCPP?

- ISO 15118: Communication between EV and charging station.
- OCPP: Communication between charging station and central management system.

Q3. What happens if the charger output current becomes too high?

The controller detects the abnormal current using sensors and protection circuits, reduces or disables power switching, and disconnects the output using the appropriate protection sequence.

7. Your Practical Learning Roadmap

I recommend learning these topics in this order:

1. Electronics fundamentals
   Voltage, current, resistance, power, diodes, MOSFETs, capacitors and inductors.
2. Power electronics
   Rectifiers, PFC, buck, boost, buck-boost and DC-DC converters.
3. Embedded systems
   STM32, ESP32, ADC, PWM, UART, SPI and CAN.
4. EV communication
   CCS2, DIN 70121, ISO 15118, PLC and OCPP.
5. Hardware testing
   PCB inspection, sensor calibration, fault detection, contactor testing and troubleshooting.
6. EV charger architecture
   Understand how all these components work together.

8. Recommended Technical Resources

These are useful references for deeper learning:

1. NITI Aayog – Fundamentals of Electric Vehicle Charging Technology. Particularly useful for understanding charger architecture.
2. Texas Instruments – EV charging controllers and communication architecture.
3. onsemi – DC Fast EV Charging: Common System Topologies and Power Devices.
4. Microchip Technology – Level 3 DC Fast Charging.

My recommendation for your career preparation

Focus particularly on these four skills:

- Embedded C and STM32
- Power electronics and converter fundamentals
- CAN communication and CCS2/PLC architecture
- PCB hardware testing and troubleshooting

One important safety point: commercial DC fast chargers operate at dangerous high voltages and currents. Learn their architecture through simulations and low-voltage training circuits; never probe a live high-voltage charger without formal training and authorized safety procedures.

One question: Are you studying this topic for an EV charger controller job interview, or do you want to learn how to actually design and develop a DC fast charger controller?

That distinction will help me prepare the right practical training path.